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A Deep Convolutional Neural Network -Based Approach For Signature Forgery Detection In Documents

Team no.3

MLA0405 – Deep Learning

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Abstract

- **Develops a Deep Convolutional Neural Network (DCNN)** to detect forged and genuine signatures automatically.
- **Preprocesses signature images** using resizing, grayscale conversion, and normalization to improve model performance.
- **Extracts important signature features automatically** without requiring manual feature engineering.
- **Classifies signatures accurately** as genuine or forged using deep learning techniques.
- **Evaluates the model** using performance metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Aim and Objective

Aim:

- To develop a **Deep Convolutional Neural Network (DCNN)-based system** for accurately detecting forged signatures in documents by automatically learning signature features and classifying signatures as genuine or forged.

Objectives:

- To collect and preprocess genuine and forged signature images for model training and testing.
- To develop a Deep Convolutional Neural Network (DCNN) for accurate classification of signatures as genuine or forged.
- To evaluate the performance of the proposed model using metrics such as Accuracy, Precision, Recall, and F1-Score.

Introduction

- Signature verification is an important process used to authenticate documents and prevent fraud in banking, legal, and business applications.
- Manual signature verification is time-consuming, subjective, and prone to human errors, making automated solutions essential.
- Deep learning techniques, especially Deep Convolutional Neural Networks (DCNNs), can automatically learn complex features from signature images.
- A DCNN-based approach improves the accuracy and reliability of distinguishing genuine signatures from forged ones.
- The proposed system preprocesses signature images and classifies them as genuine or forged using a trained deep learning model.

Literature Review & Requirement Analysis

S.No	Year of Publication	Author(s)	Title of the Journal / Book	Methodology Used	Algorithm Used	Key Findings
1	2026	Zhang et al.	IEEE Access – Deep Learning-Based Offline Signature Verification	Signature image preprocessing and deep feature extraction	DCNN	Achieved high forgery detection accuracy with reduced false positives.
2	2025	Kim et al.	Expert Systems with Applications – Signature Forgery Detection Using Deep Learning	Offline signature classification using image analysis	CNN + Transfer Learning	Improved classification performance for genuine and forged signatures.
3	2024	Li et al.	Pattern Recognition Letters	Automatic feature extraction from signature images	ResNet-50	Enhanced verification accuracy and robustness against skilled forgeries.
4	2023	Wang et al.	IEEE Transactions on Information Forensics and Security	Signature verification using deep neural networks	CNN	Produced reliable authentication with lower error rates than traditional methods.
5	2022	Chen et al.	Computers & Security	Image preprocessing and signature classification	VGG16	Improved forgery detection accuracy but required a large training dataset.

Problem Statement

- Manual signature verification is time-consuming, inconsistent, and prone to human errors, making it difficult to accurately detect forged signatures in important documents.
- Traditional signature verification methods rely on handcrafted features and often fail to identify skilled forgeries, highlighting the need for an accurate and automated Deep Convolutional Neural Network (DCNN)-based detection system.

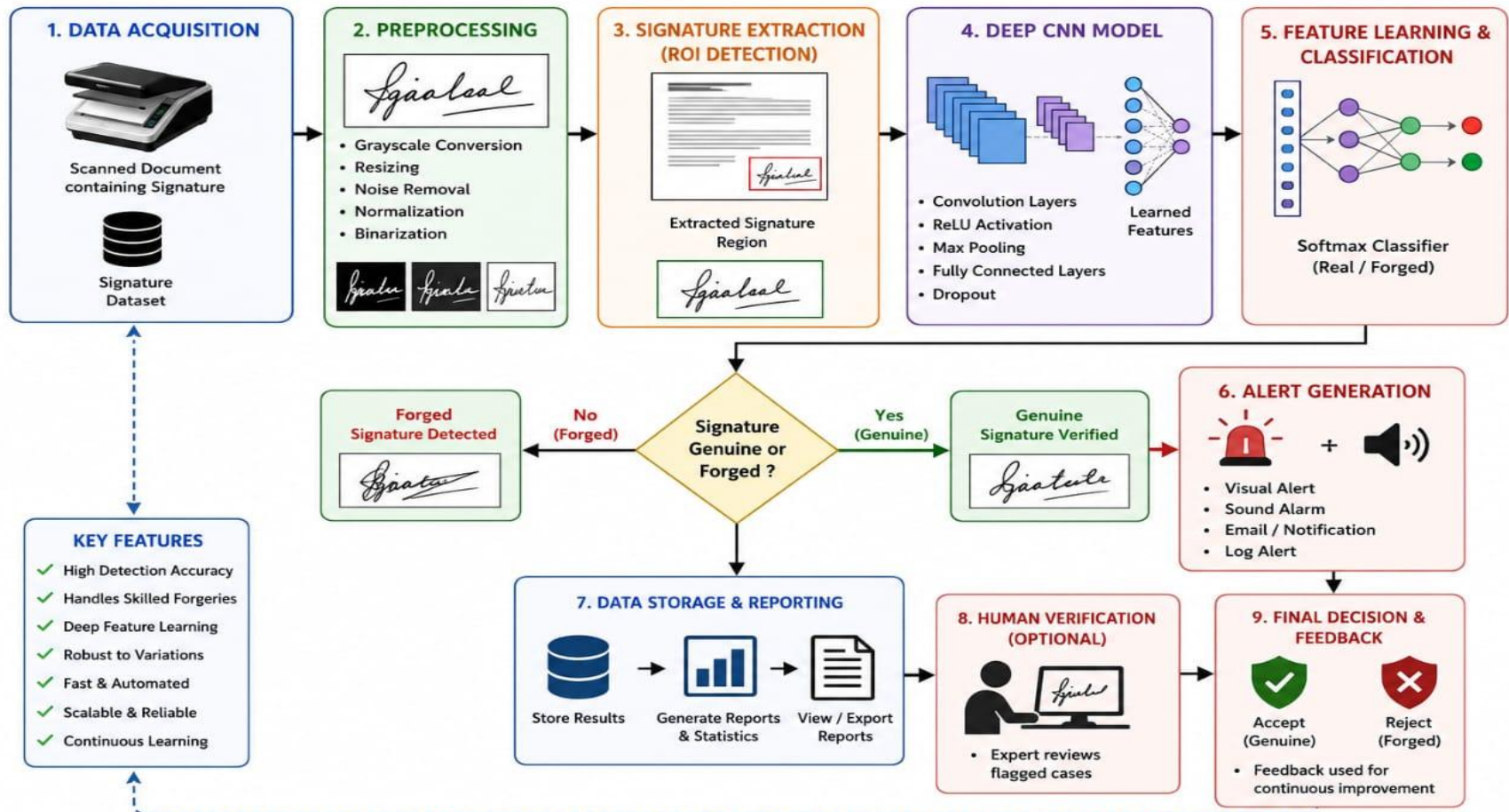
Existing System

- **Manual verification:** Signature authentication is mainly performed by human experts, making the process slow and prone to errors.
- **Traditional feature extraction:** Existing methods rely on handcrafted features, which are often ineffective in detecting skilled forgeries.
- **Limited accuracy:** Conventional machine learning techniques may fail to capture complex signature patterns, resulting in lower detection accuracy.
- **Poor scalability:** Existing systems require significant manual effort and are not suitable for processing large volumes of documents efficiently.

Proposed System

- The proposed system uses a **Deep Convolutional Neural Network (DCNN)** to automatically learn and extract important signature features from document images.
- It applies **image preprocessing techniques** such as noise removal, resizing, and normalization to improve signature analysis.
- The extracted signature patterns are analyzed by the DCNN model to differentiate between **genuine and forged signatures**.
- The system provides an **efficient and accurate approach** for automatic signature verification and forgery detection.
- It reduces human effort and improves the reliability of document authentication processes.

Overall System Architecture Design



Design & Module Planning

Module 1: Data Collection and Preprocessing

Module 2: Deep CNN-Based Signature Forgery Detection

Module 3: Performance Evaluation and Result Analysis

Conclusion

Problem: Signature forgery in documents is a major challenge that affects document security and authentication due to the limitations of traditional manual verification methods.

Solution: The proposed **DCNN-based signature forgery detection system** provides an automated approach by using image preprocessing and deep learning techniques to extract signature features and classify signatures as genuine or forged.

Impact: The system improves the accuracy, speed, and reliability of signature verification while reducing human effort in document authentication.

Significance: This project demonstrates the importance of deep learning in developing secure and efficient solutions for detecting signature fraud in digital documents.

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